MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – WEEKLY TEST 06** | **Time:** 45 Minutes
Class: Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM:** 20

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: MULTIPLE CHOICE

MATHS MANTHAN | (5 × 1 = 5 Marks)

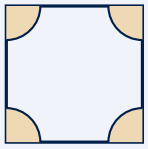
Q1. From each corner of a square of side 4 cm, a quadrant of a circle of radius 1 cm is cut. The sum of the areas of the four quadrants is: (Use $\pi = 3.14$) [1]

(A) 3.14 cm^2 (B) 6.28 cm^2 (C) 12.56 cm^2 (D) 1.57 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Sum of Four Quadrants [0.5 Mark]

Each quadrant has radius $r = 1$ cm. Four identical quadrants combine to form one complete circle:



$$\text{Sum of areas} = 4 \times \left(\frac{1}{4}\pi r^2\right) = \pi r^2$$

Step 2: Substitution [0.5 Mark]

$$\text{Sum of areas} = 3.14 \times (1)^2 = 3.14 \text{ cm}^2$$

Correct Option: (A) 3.14 cm^2

Combined Area of 4 Quadrants = **3.14** cm^2

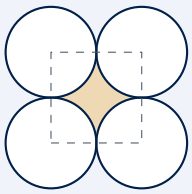
Q2. Four equal circles of radius a touch each other externally. The area enclosed between the four circles is: [1]

(A) $(4 - \pi)a^2$ (B) $(2 - \pi)a^2$ (C) $(\pi - 2)a^2$ (D) $(4 - 2\pi)a^2$

► STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Geometry of Tangent Circles [0.5 Mark]

Connecting the centres of the four mutually tangent circles forms a square of side $2a$.



$$\text{Area of square} = (2a)^2 = 4a^2$$

Step 2: Subtracting Four Quadrants [0.5 Mark]

The four sectors inside the square are quadrants of radius a, which together equal one circle of area πa^2:

$$\text{Enclosed Area} = \text{Area of square} - \pi a^2 = 4a^2 - \pi a^2 = (4-\pi)a^2$$

Correct Option: **(A)** $(4-\pi)a^2$

$$\boxed{\text{Enclosed Area} = (\mathbf{4}-\pi)\mathbf{a^2}}$$

Q3. A racetrack is in the form of a ring. If the difference between the outer and inner circumferences is 44 m, then the width of the track is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 14 m (B) 7 m (C) 3.5 m (D) 22 m

► STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Relation between Circumference Difference and Width [0.5 Mark]

Let outer radius be R and inner radius be r. Then track width $w = R - r$.

$$C_2 - C_1 = 2\pi R - 2\pi r = 2\pi(R-r) = 2\pi w$$

Step 2: Solving for Width w [0.5 Mark]

$$44 = 2 \times \frac{22}{7} \times w \implies 44 = \frac{44}{7}w \implies w = 7 \text{ m}$$

Correct Option: **(B) 7 m**

$$\boxed{\text{Width of the track } w = \mathbf{7} \text{ m}}$$

Q4. If the radius of a circle is increased by 50%, then its area is increased by: [1]

(A) 50% (B) 100% (C) 125% (D) 150%

► STEP-BY-STEP SOLUTION: [1 Mark]

Step 1: Compute New Radius and Area [0.5 Mark]

Initial radius $= r \implies$ Initial Area $A = \pi r^2$.

$$\text{New radius } r' = r + 0.50r = 1.5r$$

$$\text{New Area } A' = \pi(1.5r)^2 = 2.25\pi r^2 = 2.25A$$

Step 2: Percentage Increase in Area [0.5 Mark]

$$\text{Percentage Increase} = \frac{A' - A}{A} \times 100\% = \frac{2.25A - A}{A} \times 100\% = 1.25 \times 100\% = 125\%$$

Correct Option: **(C)** 125%

$$\boxed{\text{Percentage Increase} = \mathbf{125}\%}$$

Q5. **Assertion (A):** If a circle circumscribes a square of side a, then the area of the circle is $\frac{\pi a^2}{2}$.
Reason (R): The diagonal of the inscribed square coincides with the diameter of the circumscribing circle. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Step 1: Verifying Reason (R) [0.5 Mark]

The vertices of an inscribed square lie on the circle, and each opposite pair forms a straight line through the center of length $a\sqrt{2}$. Reason (R) is **True**.

Step 2: Verifying Assertion (A) [0.5 Mark]

Diameter $2R = a\sqrt{2} \implies R = \frac{a}{\sqrt{2}}$.

$$\text{Area of circle} = \pi R^2 = \pi \left(\frac{a}{\sqrt{2}}\right)^2 = \frac{\pi a^2}{2}$$

Thus Assertion (A) is **True**, and Reason (R) is the correct explanation.

Correct Option: **(A) Both (A) and (R) are true and (R) is the correct explanation of (A).**

Correct Option: (**A**)

SECTION B: Very Short Answer Type

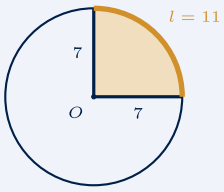
MATHS MANTHAN | **(1 × 2 = 2 Marks)**

Q6. Find the area of the sector of a circle of radius 7 cm if the length of the corresponding arc is 11 cm. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: Radius $r = 7$ cm, Arc length $l = 11$ cm.

Step 1: State Formula Relating Area to Arc Length [1 Mark]



$$\text{Area of sector} = \frac{1}{2}lr$$

Step 2: Substitution and Simplification [1 Mark]

$$\text{Area} = \frac{1}{2} \times 11 \times 7 = \frac{77}{2} = 38.5 \text{ cm}^2$$

Area of the sector = **38.5** cm^2 $\left(\text{or } \frac{\mathbf{77}}{\mathbf{2}} \text{ cm}^2\right)$

SECTION C: Short Answer Type

MATHS MANTHAN | **(3 × 3 = 9 Marks)**

Q7. A car has two windshield wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115°. Calculate the total area cleaned at each sweep of the blades. $\left(\text{Use } \pi = \frac{22}{7}\right)$ **[3]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Given: Length of blade $r = 25$ cm, Sweep angle $\theta = 115°$.



Step 1: Area Cleaned by One Blade [1.5 Marks]

$$\text{Area}_1 = \frac{\theta}{360°} \times \pi r^2 = \frac{115°}{360°} \times \frac{22}{7} \times (25)^2$$

$$= \frac{23}{72} \times \frac{22}{7} \times 625 = \frac{23 \times 11 \times 625}{36 \times 7} = \frac{158125}{252} \text{ cm}^2$$

Step 2: Total Area Cleaned by Both Blades [1.5 Marks]

Since the wipers do not overlap:

$$\text{Total Area} = 2 \times \text{Area}_1 = 2 \times \frac{158125}{252} = \frac{158125}{126} \text{ cm}^2 \approx 1254.96 \text{ cm}^2$$

$$\boxed{\text{Total Area Cleaned} = \mathbf{\frac{158125}{126}} \text{ cm}^{\mathbf{2}} \approx \mathbf{1254.96} \text{ cm}^{\mathbf{2}}}$$

Q8. A calf is tied with a rope of length 6 m at one corner of a rectangular grassy lawn of dimensions 20 m × 16 m. Find:

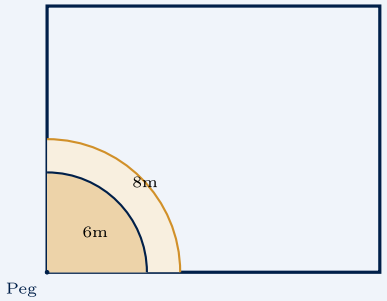
(i) the area of the lawn the calf can graze, and

(ii) the increase in grazing area if the rope length is increased to 8 m. (Use $\pi = 3.14$)

[3]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

At the corner of a rectangular lawn, the sector subtends an angle of 90° (a quadrant).



Part (i): Grazing Area with $r_1 = 6$ m Rope [1.5 Marks]

$$\text{Area}_1 = \frac{1}{4} \times \pi r_1^2 = \frac{1}{4} \times 3.14 \times (6)^2 = \frac{1}{4} \times 3.14 \times 36 = 28.26 \text{ m}^2$$

Part (ii): Increase in Grazing Area with $r_2 = 8$ m Rope [1.5 Marks]

Since 8 m < 16 m and 8 m < 20 m, the shape remains a quadrant:

$$\text{Area}_2 = \frac{1}{4} \times \pi r_2^2 = \frac{1}{4} \times 3.14 \times (8)^2 = \frac{1}{4} \times 3.14 \times 64 = 50.24 \text{ m}^2$$

$$\text{Increase in Area} = \text{Area}_2 - \text{Area}_1 = 50.24 - 28.26 = 21.98 \text{ m}^2$$

$$\boxed{\text{(i) Initial Area} = \mathbf{28.26} \text{ m}^{\mathbf{2}}, \quad \text{(ii) Increase in Area} = \mathbf{21.98} \text{ m}^{\mathbf{2}}}$$

Q9. An archery target has its gold region with diameter 21 cm and each of the outer concentric bands is 10.5 cm wide. Find:

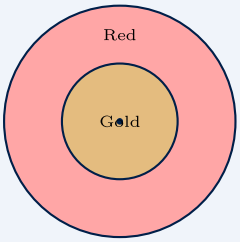
(i) the area of the gold region, and

(ii) the area of the red band. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Diameter of gold region $d_1 = 21$ cm $\implies$ $r_1 = 10.5$ cm. Radial width $w = 10.5$ cm.



Part (i): Area of Gold Region [1.5 Marks]

$$\text{Area}_{\text{Gold}} = \pi r_1^2 = \frac{22}{7} \times (10.5)^2 = \frac{22}{7} \times 110.25 = 22 \times 15.75 = 346.5 \text{ cm}^2$$

Part (ii): Area of Red Band [1.5 Marks]

Outer radius of red band $r_2 = 10.5 + 10.5 = 21$ cm.

$$\text{Area}_{\text{Red}} = \pi(r_2^2 - r_1^2) = \frac{22}{7} \times (21^2 - 10.5^2) = \frac{22}{7} \times (441 - 110.25)$$

$$= \frac{22}{7} \times 330.75 = 22 \times 47.25 = 1039.5 \text{ cm}^2$$

(i) $\text{Area}_{\text{Gold}} = \mathbf{346.5}$ cm^2, (ii) $\text{Area}_{\text{Red}} = \mathbf{1039.5}$ cm^2

SECTION D: Case-Based / Applied Situation — MATHS MANTHAN | (1 × 4 = 4 Marks)

Q10. Case Study: Pasture Grazing Dynamics on a Square Meadow

A horse is tethered with a rope to a corner peg of a square fenced grass meadow of side 15 m. Initially, the rope length is 5 m. Later, the farmer extends the rope length to 10 m to permit access to more grazing area.

(i) What geometric shape is swept by the horse while grazing within the meadow boundary? [1]

(ii) Find the initial grazing area accessible to the horse with the 5 m rope. [1]

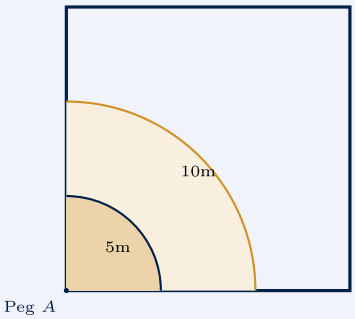
(iii) Calculate the increase in the grazing area when the rope is extended to 10 m. (Use $\pi = 3.14$) [2]

— OR —

Find the remaining ungrazed area of the meadow when the rope is 10 m long. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Given: Side of square field $a = 15$ m. Initial rope $r_1 = 5$ m, Extended rope $r_2 = 10$ m.



Part (i): Geometric Shape Swept [1 Mark]

Since the corner angle of the square meadow is 90°, the grazed region is a **quadrant of a circle** (or a circular sector with central angle 90°).

Part (ii): Initial Grazing Area ($r_1 = 5$ m) [1 Mark]

$$\text{Area}_1 = \frac{1}{4} \times \pi r_1^2 = \frac{1}{4} \times 3.14 \times 5^2 = \frac{78.5}{4} = 19.625 \text{ m}^2$$

Part (iii): Increase in Grazing Area ($r_2 = 10$ m) [2 Marks]

$$\text{Area}_2 = \frac{1}{4} \times \pi r_2^2 = \frac{1}{4} \times 3.14 \times 10^2 = \frac{314}{4} = 78.5 \text{ m}^2$$

$$\text{Increase in Area} = \text{Area}_2 - \text{Area}_1 = 78.5 - 19.625 = 58.875 \text{ m}^2$$

— OR (Alternative Choice) —

Remaining Ungrazed Area with 10 m Rope: [2 Marks]

$$\text{Total Area of Meadow} = a^2 = 15 \times 15 = 225 \text{ m}^2$$

$$\text{Remaining Ungrazed Area} = 225 - 78.5 = 146.5 \text{ m}^2$$

(i) Quadrant of a Circle, (ii) **19.625 m²**, (iii) Increase = **58.875 m²** (OR Ungrazed = **146.5 m²**)

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★